

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>

11490434/11490434 ————— **0s** 0us/step

2026-05-21 23:42:56.369631: E external/local_xla/xla/stream_executor/cuda/cuda_platform.cc:51] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)

Початок навчання моделі...

Epoch 1/10

1875/1875 ————— **9s** 4ms/step - accuracy: 0.8746 - loss: 0.4288 -
val_accuracy: 0.9540 - val_loss: 0.1522

Epoch 2/10

1875/1875 ————— **7s** 4ms/step - accuracy: 0.9670 - loss: 0.1073 -
val_accuracy: 0.9691 - val_loss: 0.0973

Epoch 3/10

1875/1875 ————— **7s** 4ms/step - accuracy: 0.9777 - loss: 0.0713 -
val_accuracy: 0.9745 - val_loss: 0.0843

Epoch 4/10

1875/1875 ————— **7s** 4ms/step - accuracy: 0.9832 - loss: 0.0525 -
val_accuracy: 0.9761 - val_loss: 0.0811

Epoch 5/10

1875/1875 ————— **7s** 4ms/step - accuracy: 0.9880 - loss: 0.0394 -
val_accuracy: 0.9769 - val_loss: 0.0747

Epoch 6/10

1875/1875 ————— **7s** 4ms/step - accuracy: 0.9894 - loss: 0.0317 -
val_accuracy: 0.9784 - val_loss: 0.0791

Epoch 7/10

1875/1875 ————— **7s** 4ms/step - accuracy: 0.9912 - loss: 0.0254 -
val_accuracy: 0.9790 - val_loss: 0.0789

Epoch 8/10

1875/1875 ————— **8s** 4ms/step - accuracy: 0.9932 - loss: 0.0204 -
val_accuracy: 0.9712 - val_loss: 0.1157

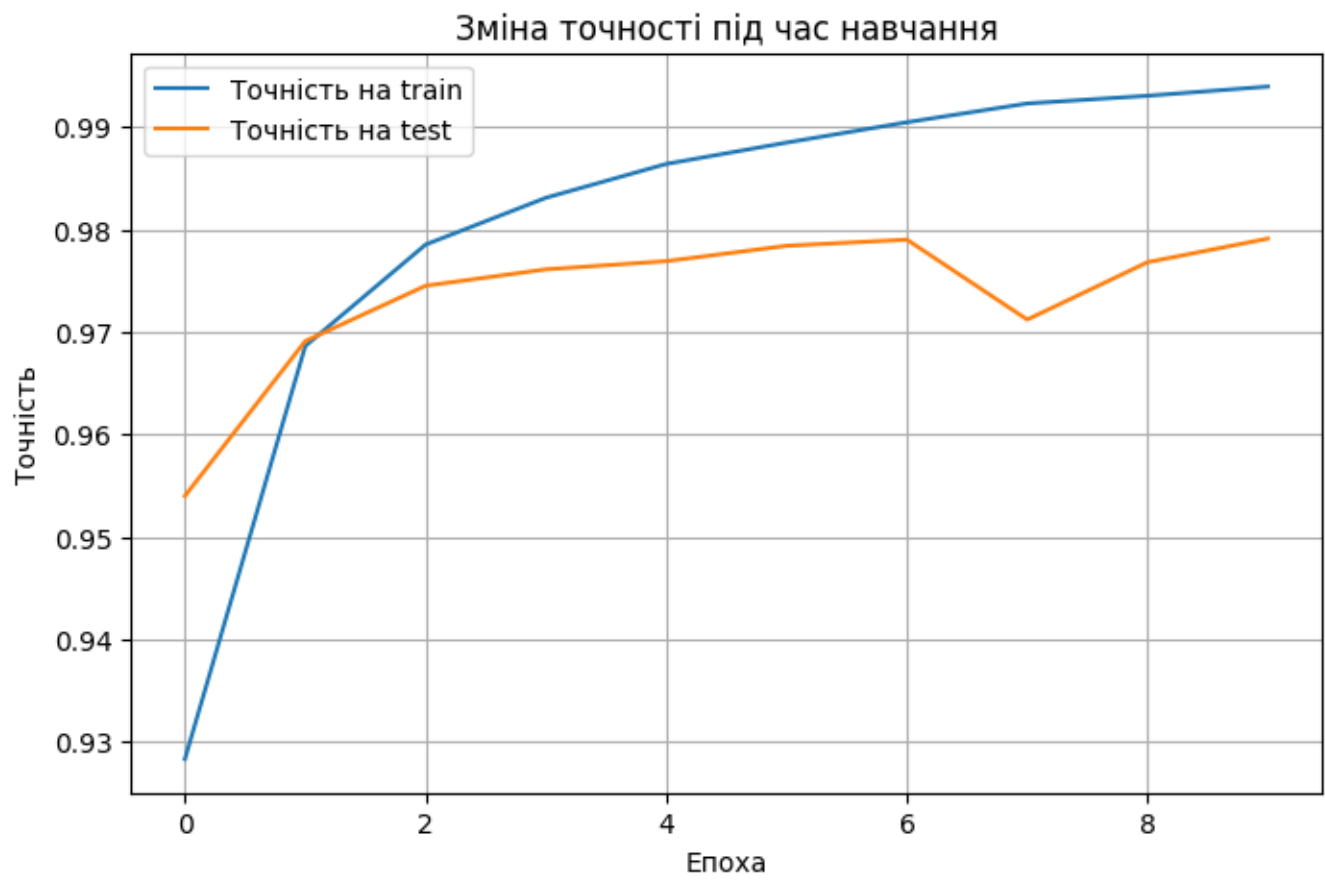
Epoch 9/10

1875/1875 ————— **7s** 4ms/step - accuracy: 0.9937 - loss: 0.0186 -
val_accuracy: 0.9768 - val_loss: 0.0876

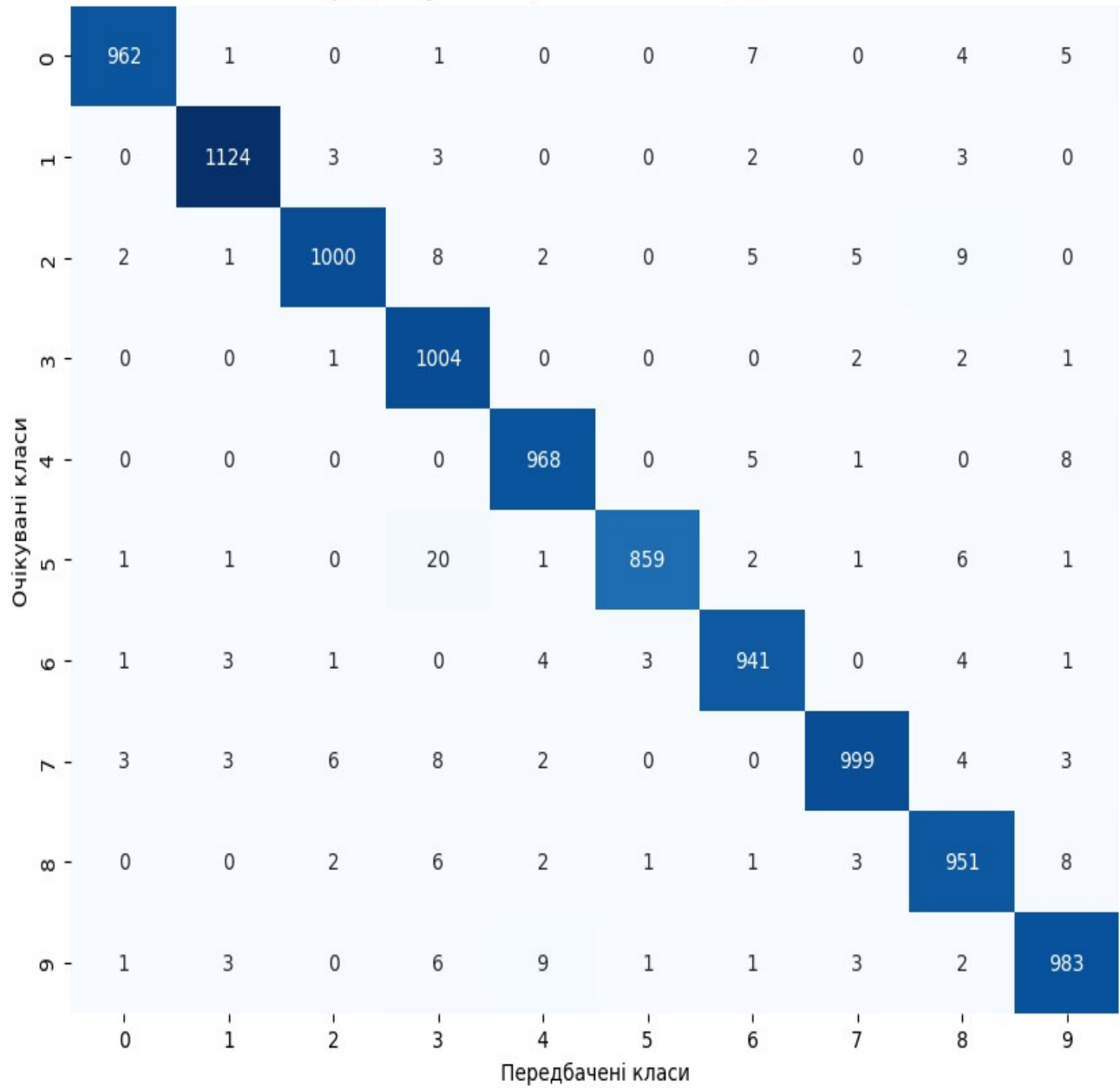
Epoch 10/10

1875/1875 ————— **7s** 4ms/step - accuracy: 0.9944 - loss: 0.0172 -
val_accuracy: 0.9791 - val_loss: 0.0942

Точність на тестових даних (Accuracy): 97.91%



Матриця плутанини (Confusion Matrix) для 10 класів



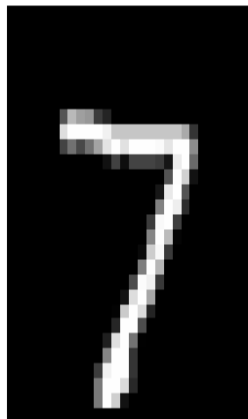
--- Звіт про класифікацію (Precision, Recall, F1-Score) ---

	precision	recall	f1-score	support
0	0.9918	0.9816	0.9867	980
1	0.9894	0.9903	0.9899	1135
2	0.9872	0.9690	0.9780	1032
3	0.9508	0.9941	0.9719	1010
4	0.9798	0.9857	0.9827	982
5	0.9942	0.9630	0.9784	892
6	0.9761	0.9823	0.9792	958
7	0.9852	0.9718	0.9785	1028

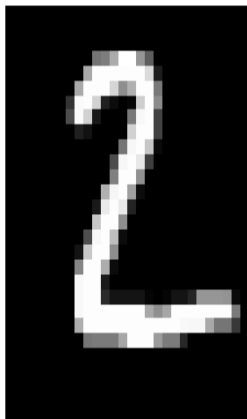
8	0.9655	0.9764	0.9709	974
9	0.9733	0.9742	0.9737	1009

accuracy		0.9791	10000
macro avg	0.9793	0.9788	0.9790
weighted avg	0.9793	0.9791	0.9791

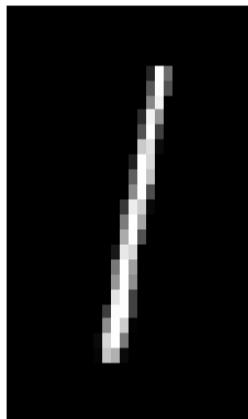
Передбачено: 7
Справжня: 7



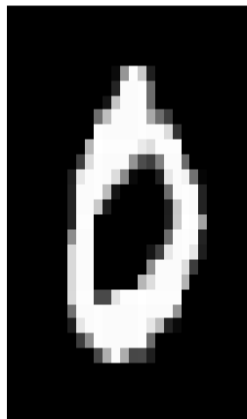
Передбачено: 2
Справжня: 2



Передбачено: 1
Справжня: 1



Передбачено: 0
Справжня: 0



Передбачено: 4
Справжня: 4

